SwinMatcher: Universal Cross-Modal Remote Sensing Image Matching With Interactive Swin Transformer

Wei Li, *Senior Member, IEEE*, Desheng Weng, Chenzhong Gao, *Graduate Student Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*—Cross-modal remote sensing image matching serves as a key technique for collaborative utilization of multisource information. However, modal differences and geometric distortions between multisource images pose challenges to existing methods in terms of robustness and generalization. To achieve feature interaction in cross-modal scenes, this article proposes SwinMatcher, an end-to-end matching model based on the Transformer architecture. Innovatively proposing the window/shifted-window cross-attention (SWCA) based on the window/shifted-window self-attention (SWSA) mechanisms of Swin Transformer, SwinMatcher enables efficient cross-modal feature interaction and multiscale contextual modeling. It also incorporates a learnable matching module to directly generate semidense correspondences. Moreover, a cross-modal remote sensing image matching dataset is generated, which encompasses four modalities: visible light, synthetic aperture radar (SAR), light detection and ranging (LiDAR), and map, distributed across four representative scenes. The dataset includes 400 samples produced via random homography transformations, designed to enhance modal diversity and scene complexity. Experiments demonstrate that SwinMatcher outperforms state-of-the-art methods on this new dataset as well as public benchmarks, exhibiting superior robustness under complex scenes involving coupled modal and geometric distortions. The proposed method and dataset provide novel solutions and evaluation benchmarks for cross-modal remote sensing image matching. The code and testing dataset will be made publicly available at https://github.com/LotrL/SwinMatcher**

***Index Terms*—Cross-modal, deep learning, image matching, multisource, remote sensing, Swin Transformer.**

I. Introduction

CROSS-MODAL remote sensing image matching is a core technique for collaborative multisource remote sensing data analysis, playing a pivotal role in image fusion [1], [2], change detection [3], [4], and precise navigation [5]. With the rapid development of sensor technologies, heterogeneous remote sensing data, such as optical, synthetic aperture radar (SAR), light detection and ranging (LiDAR), and multitemporal sources, have experienced exponential growth. Their complementary nature offers multidimensional perspectives for dynamic Earth observation and complex scene understanding. Cross-modal matching overcomes the limitations of single-modal perception by integrating information from diverse modalities, facilitating reliable analysis in challenging scenes. However, effectively integrating complementary advantages of different modalities while addressing the challenges posed by their varying imaging mechanisms remains a critical issue [6], [7], [8].

The challenges of cross-modal remote sensing image matching arise from the coupling effects of modal discrepancies and scene complexity. First, radiometric differences coexist with geometric distortions. Significant nonlinear radiometric variations arise from distinct imaging principles, while viewpoint transition and terrain undulations lead to geometric inconsistencies such as scale changes, rotation, and nonrigid deformations. Second, feature representation mismatch persists. Traditional handcrafted features rely on shallow gradients or phase information, and they fail to capture structural commonalities across different modalities [9], [10], [11]. Deep learning models, despite their ability to extract high-level semantic features, still struggle with limited generalization in cross-modal matching due to an insufficient feature perception mechanism and a lack of annotated data. Recent studies [12], [13] train end-to-end models on specific modality pairs, such as visible-infrared or visible-SAR, to learn deep semantic representations. However, their overfitting to a narrow range of modalities limits generalization to others, such as digital maps or nighttime images. Moreover, performance heavily relies on the quality and quantity of annotated data. Designing methods that balance computational efficiency, cross-modal generalization, and robustness against geometric distortions remains a core challenge.

To address these issues, we collect multimodal remote sensing images from diverse scenes to train and test models, enabling robust learning of invariant features across modalities and alleviating benchmark scarcity. Inspired by the success of Swin Transformer [14], [15], [16], we integrate its

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Wei Li, Desheng Weng, and Chenzhong Gao are with Beijing Institute of Technology, Zhuhai, Guangdong 519088, China, and also with the School of Information and Electronics and the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing Institute of Technology, Beijing 100081, China (e-mail: liwei089@ieee.org; lotrl@qq.com; gao-pingqi@qq.com).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

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shifted-window mechanisms with cross-modal interaction for feature extraction to further explore its application potential. This architecture captures intra-modal structural consistency and intermodal semantic complementarity, and it integrates a hierarchical matching module to achieve end-to-end cross-modal image matching.

The main contributions of this article are summarized as follows.

1) Interactive Swin Transformer Blocks (ISTBs) are proposed, which innovatively introduce the shifted-window cross-attention (SWCA) based on the feature interaction mechanism of Swin Transformer. Through a multiscale window partition strategy, it establishes a local–global collaborative feature interaction paradigm, enabling the extraction of invariant features across diverse modality images and improving adaptability to complex geometric distortions.
2) A coarse-to-fine hierarchical matching pipeline is designed, innovatively employing a convolutional block that captures local features to achieve subpixel offset estimation. This approach not only increases the upper limit of matching quantities but also enhances the localization accuracy in cross-modal image matching.
3) A cross-modal dataset covering visible light, SAR, LiDAR, and Map modalities is presented as a new evaluation benchmark, consisting of 400 samples generated using random homography. Through a stepwise training strategy and effective data augmentation methods, the proposed SwinMatcher demonstrates state-of-the-art performance on both the new dataset and public benchmarks, particularly exhibiting excellent matching robustness in scenes with significant modal differences and complex geometric distortions.

The remainder of this article is organized as follows: Section II reviews related work. Section III details the proposed methodology. Section IV presents experimental results and analysis. Section V concludes this article.

II. Related Work

A. Traditional Handcrafted Matching Methods

Traditional handcrafted cross-modal image matching methods have evolved into two main categories: area-based and feature-based approaches.

Area-Based: Matching methods align images by calculating similarity metrics between corresponding local regions. Typical metrics include normalized cross correlation (NCC) [17], [18], sum of squared differences (SSDs), and mutual information (MI) [19]. Among these, MI exhibits robustness to nonlinear radiometric differences, making it widely adopted in cross-modal matching [20], [21]. To further enhance cross-modal robustness, researchers have proposed structure-based improvements, such as HOPC [22] and CFOG [23], which mitigate modal differences by extracting structural or phase information. However, these methods rely on sliding-window strategies for global search, leading to high-computational complexity and susceptibility to local optima [21]. Moreover, their adaptability to geometric distortions is limited, especially in complex scenes.

Feature-Based: Matching methods align images by detecting keypoints, constructing descriptors, and performing feature matching. Classic methods such as SIFT [24] and SURF [25] build descriptors using gradient information but struggle to handle cross-modal radiometric variations. To address this limitation, researchers develop improvements targeting cross-modal challenges. For instance, RIFT [9] employs maximum index maps to extract phase congruency features for radiometric invariance. LNIFT [10] introduces local normalization to improve computational efficiency. MS-HLMO [26] fuses multiscale local main orientation histograms with phase and gradient features to enhance feature distinctiveness in complex scenes. SRIF [27] combines projection pyramid scale space and local intensity binarization transformation to achieve rotation and scale invariance. Despite their success on specific datasets, these methods rely on handcrafted feature extractors, which struggle to fully exploit cross-modal commonalities. In scenes with repetitive textures, keypoint detection stability declines, and descriptors exhibit limited robustness to viewpoint or perspective changes.

B. Deep Learning-Based Matching Methods

Recent advancements in deep learning for image matching can be broadly categorized into detector-based and detector-free approaches.

Detector-Based: Methods follow the traditional feature-matching pipeline, comprising keypoint detection, feature description, and matching. The representative SuperPoint [28] generates keypoints and descriptors via self-supervised learning to enhance generalization through synthetic and real data. Subsequent works [29], [30], [31] optimize detection and description processes, particularly D2-Net [32], which unifies both by defining keypoints as channel-wise maxima in feature maps, thereby improving robustness to viewpoint changes. Graph-based methods like Superglue [33] model matching as a graph neural network problem, leveraging self-attention and cross-attention for global information propagation and formulating correspondence estimation as an optimal transport problem. LightGlue [34] optimizes Superglue with adaptive depth halting, width pruning, lightweight prediction heads, and rotary positional encoding, achieving efficient and accurate sparse matching. While effective in texture-rich scenes, these methods heavily depend on detector performance and struggle with extreme viewpoints or textureless regions.

Detector-Free: Methods bypass explicit keypoint detection, directly inferring matches from dense features, and often exploit global attention to overcome local receptive field limitations. The typical LoFTR [35] generates semidense matches in textureless regions using alternating self-attention and cross-attention modules. Subsequent works [36], [37], [38] built on this foundation to improve matching performance, particularly the Efficient LoFTR [39], which addresses computational redundancy in LoFTR via aggregated attention, reducing complexity through adaptive feature aggregation and local attention. Additionally, XoFTR [13] tackles visible-infrared

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cross-modal matching with masked image modeling pretraining, pseudo-thermal image augmentation, and multiscale bidirectional attention. For dense matching, RoMa [40] combines DINOv2 [41] pretrained features with CNN fine-level features, employing a Transformer-based anchor probability prediction decoder and a hybrid classification-regression loss to achieve robust matching results. Detector-free methods leverage end-to-end learning for contextual awareness, excelling in cross-modal and large-displacement scenes, but still leave room for improvement and require a balance between accuracy and efficiency.

C. Deep Learning for Cross-Modal Remote Sensing Image Matching

Deep-learning methods for cross-modal remote sensing image matching increasingly leverage learnable feature encoders and global descriptors. Siamese-style networks remain common, as CRS-Net [42] adopts a two-branch Siamese encoder with multiscale feature extraction and a cluster-aware contrastive loss, and PCNet [43] uses a shared-weight CNN to capture structural similarities across modalities. Other pipelines break the task into detection and matching, with Li et al. [44] designing a self-supervised keypoint detector followed by a cross-fusion matching network, yielding highly repeatable keypoints in optical–SAR images. Transformer-based descriptors are also adapted, often incorporating attention and fusion modules to integrate multiple modalities. CAM-Net [45] injects co-attention between optical–SAR feature maps, and CM-Net [46] uses contextual self-attention and cross-fusion layers to generate discriminative cross-modal descriptors. These modern Siamese, contrastive, and transformer-based frameworks significantly outperform traditional handcrafted features under cross-modal conditions.

Other recent studies employ domain adaptation and graph-based matching methods to bridge modality gaps. Generative strategies translate one modality into another to simplify matching, as Hu et al. [47] use a Restormer [48] network to synthesize "pseudo-SAR" images from RGB data before registration. Likewise, vision transformers and graph neural modules originally developed for RGB matching have been applied to multisensor data, with Leahy et al. [49] showing that Superglue and LoFTR can robustly register 2-D camera imagery with LiDAR-derived feature maps. Overall, these end-to-end deep learning methods, which combine adversarial and domain-transfer components, attention layers, and global optimization, achieve far more accurate and invariant matching across optical, SAR, LiDAR, and other sensor modalities than traditional methods.

However, current research still faces challenges such as dataset scarcity and limited generalization. Building representative cross-modal datasets and developing efficient, generalizable architectures remain critical for advancing this field.

III. Methodology

The proposed framework consists of a Swin Transformer-based feature extraction module and a three-stage hierarchical feature matching architecture, as illustrated in Fig. 1. Features at different resolutions are extracted from images using a Swin Transformer-based backbone and applied at various stages of feature matching, progressively achieving coarse-to-fine matching.

A. Swin Transformer-Based Feature Extraction

Given a pair of images I_A and I_B of the same size, with height H and width W, grayscale conversion is applied to both images, followed by batch-wise concatenation. This operation aims to extract shared structural information across modalities while reducing noise interference from color channels.

1) Multiscale Feature Encoding: The input image undergoes normalization processing and enters the model at its original dimension $H \times W$. The first critical operation is patch embedding, which transforms the original pixel space into semantic feature patches. Using a 2-D convolutional layer with a kernel size of 4×4 and a stride of 4, the input image is divided into uniform grid patches. Each patch is mapped to a high-dimensional vector, generating the feature map $z_{\text{enc}}^{(0)} \in \mathbb{R}^{(H/4)\times(W/4)\times C}$, where $C = 96$ denotes the initial channel dimension. This operation compresses the resolution by a scale of 4 and extends single-channel grayscale information into multichannel semantic features, achieving a mapping from a low-dimensional pixel space to a high-dimensional feature space. The patch size $P = 4$ balances computational efficiency and local detail preservation, ensuring each patch covers sufficient contextual information while avoiding feature redundancy.

The encoder adopts a four-stage pyramid structure. At each stage, the proposed ISTBs capture contextual information from images. Through patch merging, spatial resolution is gradually reduced, and channel dimensions are expanded, yielding a multiscale hierarchical feature representation. Let the input resolution of the lth stage be $H_l \times W_l = (H/4 \times 2^{l-1}) \times (W/4 \times 2^{l-1})$ with a channel count $C_l = 96 \times 2^{l-1}$, where $l \in \{1, 2, 3, 4\}$. $S_w = 16$ represents the fixed window size. This progressive design employs a coordinated strategy of halving resolution, doubling channel dimension, and expanding windows to global scales, enabling hierarchical feature abstraction from local details to global semantics.

The patch merging operation achieves resolution halving and channel doubling by concatenating adjacent 2×2 regions, followed by linear projection. Specifically, for $\forall l \in \{2, 3, 4\}$, the input feature map $z_{\text{enc}}^{(l-1)} \in \mathbb{R}^{H_{l-1}\times W_{l-1}\times C_{l-1}}$ is sampled into four subfeatures $\{z_0^{(l-1)}, z_1^{(l-1)}, z_2^{(l-1)}, z_3^{(l-1)}\}$ with a stride of 2, each having a resolution of $(H_{l-1}/2) \times (W_{l-1}/2)$. These subfeatures are concatenated along the channel dimension and then reduce the channel count from $4C_{l-1}$ to $2C_{l-1}$ by linear projection, producing the next-stage feature map

$$z_{\text{enc}}^{(l)} = \text{Linear}\left(\text{Concat}\left(z_0^{(l-1)}, z_1^{(l-1)}, z_2^{(l-1)}, z_3^{(l-1)}\right)\right) \quad (1)$$

where $z_{\text{enc}}^{(l)} \in \mathbb{R}^{(H_{l-1}/2)\times(W_{l-1}/2)\times 2C_{l-1}}$. This operation constructs multiscale features through progressive downsampling while expanding channel dimensions to enhance semantic representation, forming a hierarchical feature pyramid.

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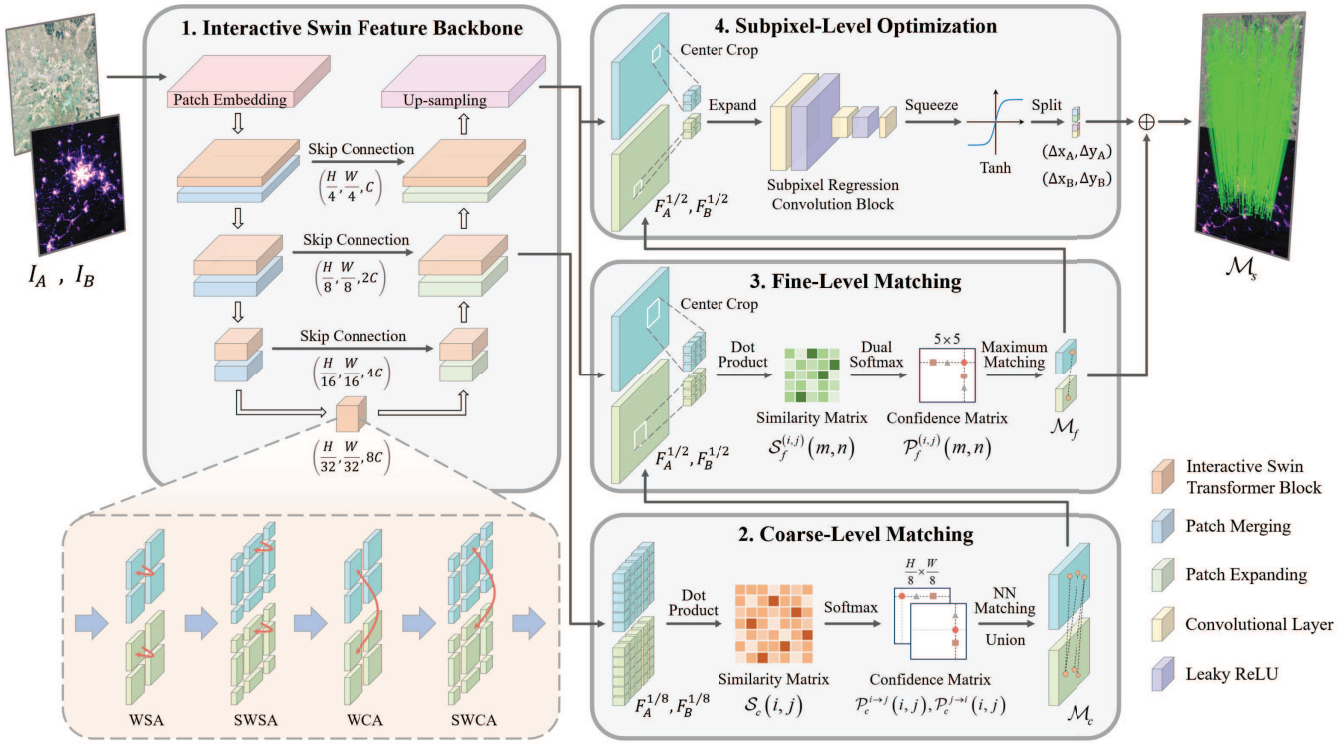


Fig. 1. Overview of the proposed method. SwinMatcher consists of a feature extraction module and a three-stage coarse-to-fine feature matching architecture: 1) the Swin Transformer feature extraction module hierarchically extracts multiscale features via a Swin Transformer-based backbone, integrating interactive window attention and cross-layer fusion to generate coarse-level and fine-level features for subsequent matching; 2) the coarse-level matching module identifies initial candidate correspondences between images through inner-product computation and confidence-based filtering, allowing one-to-many/many-to-one correspondences to increase match quantity; 3) the fine-level matching module refines correspondences within local windows by applying the bidirectional Softmax to screen the highest confidence matches, thereby adjusting positional residuals to enhancing accuracy; and 4) the subpixel-level optimization module predicts subpixel-level offsets and superimposes them onto the refined matches, surpassing resolution limitations to achieve high-precision one-to-one correspondences.

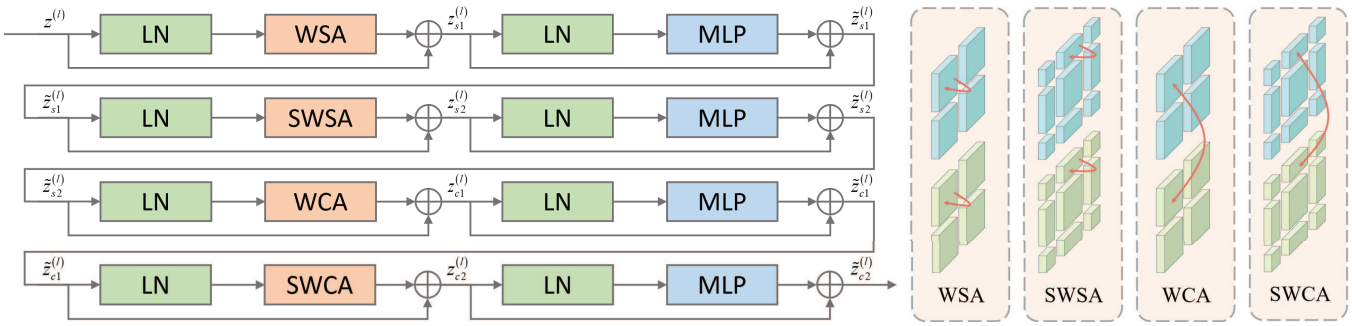


Fig. 2. ISTBs. These consist of four consecutive Swin Transformer Blocks incorporating distinct types of attention mechanisms, in which LN denotes layer normalization and MLP stands for a multilayer perceptron.

2) Interactive Swin Transformer Blocks: The proposed ISTB alternately combines window self-attention (WSA), shifted-WSA (SWSA), window cross-attention (WCA), and SWCA to construct a multilevel local–global feature interaction network. As shown in Fig. 2, each stage contains four consecutive Swin Transformer Blocks (each row represents one block). By arranging different types of attention mechanisms, the block progressively models features from local details to global semantics. Taking the first-row Swin Transformer Block as an example. Let the input feature be $z^{(l)}$. The computation is formulated as

$$z_{s1}^{(l)} = \mathrm{WSA}\left(\mathrm{LN}\left(z^{(l)}\right)\right) + z^{(l)} \tag{2}$$

$$\tilde{z}_{s1}^{(l)} = \mathrm{MLP}\left(\mathrm{LN}\left(z_{s1}^{(l)}\right)\right) + z_{s1}^{(l)} \tag{3}$$

where LN denotes layer normalization, and MLP stands for a multilayer perceptron. By sequentially replacing WSA with SWSA, WCA, and SWCA across four iterations, the final output $\tilde{z}_{c2}^{(l)}$ of the ISTB is obtained.

Window attention divides the feature map into nonoverlapping windows of size $S_w \times S_w$ and computes token correlations within each window. Shifted-window attention cyclically shifts window boundaries by $(S_w/2)$ pixels and re-partitions windows to cover noncontiguous regions in the original feature map. This asymmetric interaction design enhances the model's ability to capture long-range dependencies, mitigates the

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limited receptive field of local windows, and facilitates cross-window information exchange.

For attention computation, within each window, the input feature $z^{(l)}$ undergoes layer normalization, followed by linear transformations to generate query, key, and value matrices $Q, K, V \in \mathbb{R}^{M_l \times N_l \times S_w^2 \times d_h}$, where $M_l = (H_l/S_w) \times (W_l/S_w)$ is the number of windows, $d_h = (C_l/N_l)$ is the dimension per head, and N_l is the number of attention heads. We fix $d_h = 24$ and adjust N_l at each stage. In shifted-window attention, a mask is strategically designed to preserve window locality. A learnable relative positional bias $B_{\text{rel}} \in \mathbb{R}^{S_w^2 \times S_w^2}$ is introduced to enhance spatial position awareness. Finally, the weighted aggregation of value vectors restores the window arrangement, yielding the output $\tilde{z}^{(l)} \in \mathbb{R}^{M_l \times S_w^2 \times C_l}$.

To support subsequent matching tasks, the ISTB incorporates both self-attention and cross-attention mechanisms. Self-attention computes correlation weights among internal elements of a sequence to dynamically aggregate contextual information, capturing feature correspondences within a single image. Cross-attention, on the other hand, establishes interactions between two distinct sequences, capturing feature correspondences across two images. The two attention mechanisms are formulated as

$$\text{SA}\left(Q_{\text{A}}, K_{\text{A}}, V_{\text{A}}\right) = \text{Softmax}\left(\frac{Q_{\text{A}} K_{\text{A}}^T}{\sqrt{d_h}} + B_{\text{rel}}\right) V_{\text{A}} \tag{4}$$

$$\text{CA}\left(Q_{\text{A}}, K_{\text{B}}, V_{\text{B}}\right) = \text{Softmax}\left(\frac{Q_{\text{A}} K_{\text{B}}^T}{\sqrt{d_h}} + B_{\text{rel}}\right) V_{\text{B}} \tag{5}$$

where SA denotes self-attention, and CA denotes cross-attention. In self-attention, query, key, and value are generated from the same image. In cross-attention, the query is derived from one image, while key and value originate from the other. By integrating the proposed ISTB, the model enhances feature expressiveness, thereby improving interimage feature relevance and matching robustness.

3) Cross-Scale Feature Decoding: The decoder comprises three stages. At each stage, the features output by the ISTB from the previous stage are progressively upsampled through patch expanding and skip connections, while fusing multiscale encoder features to replenish fine details. Let the output resolution of the lth decoding stage be $H_l^{\text{up}} \times W_l^{\text{up}} = ((H/32) \times 2^l) \times ((W/32) \times 2^l)$ with a channel count $C_l^{\text{up}} = (768/2^l)$, where $l \in \{1, 2, 3\}$.

Patch expanding employs linear projection and pixel shuffle to halve the channel dimension of the output feature $\tilde{z}_{\text{dec}}^{(l-1)} \in \mathbb{R}^{H_{l-1}^{\text{up}} \times W_{l-1}^{\text{up}} \times C_{l-1}^{\text{up}}}$ (where $\tilde{z}_{\text{dec}}^{(0)} = z_{\text{enc}}^{(4)}$) from the previous ISTB while achieving $2\times$ spatial upsampling

$$z_{\text{dec}}^{(l)} = \text{PixelShuffle}\left(\text{Linear}\left(\tilde{z}_{\text{dec}}^{(l-1)}\right)\right) \tag{6}$$

where the PixelShuffle operation rearranges the linearly projected feature with $2C_{l-1}^{\text{up}}$ channels by grouping adjacent 4 channels (corresponding to a 2×2 spatial block) into one spatial pixel. This process increases the resolution by $2\times$ and reduces the channel dimension to $(C_{l-1}^{\text{up}}/2)$.

Skip connection concatenates the patch-expanded feature $z_{\text{dec}}^{(l)} \in \mathbb{R}^{2H_{l-1}^{\text{up}} \times 2W_{l-1}^{\text{up}} \times (C_{l-1}^{\text{up}}/2)}$ with the corresponding encoder feature $z_{\text{enc}}^{(4-l)}$ along the channel dimension, resulting in a combined channel count of $C_{\text{concat}} = C_l^{\text{up}} + C_{\text{enc}}^{(4-l)}$. A linear projection then aligns the channel dimensions to C_l^{up}, producing the fused feature

$$z_{\text{fused}}^{(l)} = \text{LN}\left(\text{Linear}\left(\text{Concat}\left(z_{\text{dec}}^{(l)}, z_{\text{enc}}^{(4-l)}\right)\right)\right) \tag{7}$$

where LN represents layer normalization. This strategy enhances the representational capability of the decoded features by reusing semantic information from different encoder levels. The fused feature $z_{\text{fused}}^{(l)}$ is further processed through the ISTB to generate the decoder feature $\tilde{z}_{\text{dec}}^{(l)}$.

a) Coarse-level feature: z_c. The second-stage decoder feature $z_{\text{dec}}^{(2)} \in \mathbb{R}^{(H/8) \times (W/8) \times 192}$ and the second-stage encoder feature $z_{\text{enc}}^{(2)}$ are concatenated along the channel dimension. A linear projection compresses the channels, followed by layer normalization to produce the coarse-level feature:

$$z_c = \text{LN}\left(\text{Linear}\left(\text{Concat}\left(z_{\text{dec}}^{(2)}, z_{\text{enc}}^{(2)}\right)\right)\right) \tag{8}$$

where $z_c \in \mathbb{R}^{(H/8) \times (W/8) \times 192}$ aggregates midlevel semantic information for subsequent coarse matching of feature points.

b) Fine-level feature: z_f. The third-stage decoder feature $z_{\text{dec}}^{(3)} \in \mathbb{R}^{(H/4) \times (W/4) \times 96}$ is processed through the ISTB to obtain $\tilde{z}_{\text{dec}}^{(3)}$. After layer normalization, the feature is further upsampled to twice the spatial resolution via bilinear interpolation, followed by another layer normalization:

$$z_f = \text{LN}\left(\text{Interpolate}\left(\text{LN}\left(\tilde{z}_{\text{dec}}^{(3)}\right)\right)\right). \tag{9}$$

The fine-level feature $z_f \in \mathbb{R}^{(H/2) \times (W/2) \times 96}$ retains fine-grained texture details from the original image, enabling refinement of coarsely matched feature points.

The encoder constructs a hierarchical feature pyramid through multistage downsampling and ISTB, while the decoder reconstructs features via progressive upsampling and cross-layer fusion. The dual-branch output structure (z_c and z_f) leverages complementary features to enhance overall performance. The multiscale features z_c and z_f extracted by the Swin Transformer backbone are split along the batch dimension, yielding coarse-level features $F_{\text{A}}^{1/8}$ and $F_{\text{B}}^{1/8}$ at (1/8) resolution, as well as fine-level features $F_{\text{A}}^{1/2}$ and $F_{\text{B}}^{1/2}$ at (1/2) resolution for images I_{A} and I_{B}, respectively.

B. Coarse-to-Fine Feature Matching

A three-stage hierarchical matching strategy is proposed, including coarse-level matching, fine-level matching, and subpixel-level optimization, to achieve high-precision image feature correspondence estimation. The design of each stage is elaborated below.

1) Coarse-Level Matching: For the coarse-level features $F_{\text{A}}^{1/8}, F_{\text{B}}^{1/8} \in \mathbb{R}^{(H/8) \times (W/8) \times 192}$ extracted by the backbone network, they are rearranged into a sequential format. Each element in the sequence is a 192-D feature vector. After channel normalization, pairwise inner product operations are performed between elements from two sequences to compute the similarity between the two feature sets

$$\mathcal{S}_c(i, j) = \frac{1}{\tau}\left\langle \frac{F_{\text{A}}^{1/8}(i)}{d_c}, \frac{F_{\text{B}}^{1/8}(j)}{d_c} \right\rangle \tag{10}$$

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where $\langle\cdot\rangle$ denotes the inner-product operation, i and j represent the indices of the two sequences. d_c denotes the coarse-level channel normalization coefficient, which is set to $(192)^{1/2}$. Here τ is the temperature coefficient, which regulates the smoothness of the similarity distribution. Subsequently, the Softmax function is applied along both directions of the feature sequence to obtain two directional confidence matrices

$$\mathcal{P}_c^{i\to j}(i,j) = \text{Softmax}(\mathcal{S}_c(i,\cdot))_j \tag{11}$$

$$\mathcal{P}_c^{j\to i}(i,j) = \text{Softmax}(\mathcal{S}_c(\cdot,j))_i. \tag{12}$$

The nearest neighbor (NN) matching strategy is adopted. For $\mathcal{P}_c^{i\to j}(i,j)$, the two points with the highest confidence in the jth direction are selected as candidate matches. Similarly, for $\mathcal{P}_c^{j\to i}(i,j)$, the two points with the highest confidence in the ith direction are chosen as candidate matches. A preset threshold θ_c is used to binarize the candidate matches, filtering out matches with confidence values below the threshold.

The filtered matches from both confidence matrices are merged into a union, temporarily retaining one-to-many/many-to-one matches to maximize the upper limit of matching quantity. Ultimately, the coarse-level matching point sets between the two images are obtained

$$\begin{aligned}\mathcal{M}_c = &\left\{(i,j)\,|\mathcal{P}_c^{i\to j}(i,j) \in \underset{j}{\text{NN}}\left(\mathcal{P}_c^{i\to j}(i,j)\right), \mathcal{P}_c^{i\to j}(i,j) > \theta_c\right\} \cup \\ &\left\{(i,j)\,|\mathcal{P}_c^{i\to j}(i,j) \in \underset{i}{\text{NN}}\left(\mathcal{P}_c^{j\to i}(i,j)\right), \mathcal{P}_c^{j\to i}(i,j) > \theta_c\right\}.\end{aligned} \tag{13}$$

2) Fine-Level Matching: Building upon the coarse-level matches, the fine-level matching stage aims to refine the correspondence results to enhance positional accuracy. Taking the coarse-level matched positions (i,j) as centers, an "unfolding" operation is applied to extract local window sets W_f^i and W_f^j from the fine-level feature maps $F_{\text{A}}^{1/2}, F_{\text{B}}^{1/2} \in \mathbb{R}^{(H/2)\times(W/2)\times 96}$, with a uniform window size of 5×5. Similar to the coarse-level stage, all pairwise corresponding windows are flattened and channel-normalized. A point-wise similarity matrix is then computed between each pair of windows

$$\mathcal{S}_f^{(i,j)}(m,n) = \frac{1}{\tau}\left\langle \frac{W_f^i(m)}{d_f}, \frac{W_f^j(n)}{d_f}\right\rangle \tag{14}$$

where m and n denote the indices of the two window sequences, and d_f represents the fine-level channel normalization coefficient, which is set to $(96)^{1/2}$. Unlike the coarse-level stage, a bidirectional Softmax operation (DualSoftmax) is applied to the similarity matrix. Specifically, the confidence matrices from both directions are multiplied element-wise to generate a joint confidence matrix

$$\begin{aligned}&\mathcal{P}_f^{(i,j)}(m,n) \\ &= \text{Softmax}\left(\mathcal{S}_f^{(i,j)}(m,\cdot)\right)_n \odot \text{Softmax}\left(\mathcal{S}_f^{(i,j)}(\cdot,n)\right)_m\end{aligned} \tag{15}$$

where $\odot$ indicates element-wise multiplication. Subsequently, only the pair of points with the highest confidence in $\mathcal{P}_f^{(i,j)}(m,n)$ is retained as a candidate match for each window pair. Matches with confidence values below the threshold θ_f are filtered out.

The fine-level matching stage adjusts the initially matched window center positions within their respective windows to reduce localization errors. This process resolves most one-to-many/many-to-one matches from the coarse stage into one-to-one correspondences, though a small portion may remain unresolved. Ultimately, the refined matching point sets between the two images are obtained

$$\begin{aligned}\mathcal{M}_f = \Big\{&(m,n)\,|\mathcal{P}_f^{(i,j)}(m,n) = \text{Max}\left(\mathcal{P}_f^{(i,j)}(m,n)\right), \\ &\mathcal{P}_f^{(i,j)}(m,n) > \theta_f, (i,j) \in \mathcal{M}_c\Big\}.\end{aligned} \tag{16}$$

3) Subpixel-Level Optimization: After obtaining discrete matching results within local windows from the previous stage, a subpixel regression convolution block is designed to predict positional residuals, thereby overcoming the resolution limitations of feature maps. Specifically, centered on the matched points from the previous stage, neighborhood sets W_s^m and W_s^n are extracted from the fine-level feature maps $F_{\text{A}}^{1/2}$ and $F_{\text{B}}^{1/2}$ with 3×3 windows. Each pair of feature windows is concatenated along the channel dimension to form composite features $W_s^{(m,n)} \in \mathbb{R}^{3\times 3\times 192}$. These composite features are processed through two 3×3 convolutional layers to fuse local information, yielding point-wise features $\tilde{W}_s^{(m,n)} \in \mathbb{R}^{1\times 1\times 192}$. Subsequently, a 1×1 convolutional layer adjusts the channel dimensions and splits the output into two branches. All adjacent convolutional layers incorporate LeakyReLU activation. Finally, the output of the subpixel regression convolution block is activated by the Tanh function, mapping values to the interval $(-1,1)$. This generates subpixel offsets $(\Delta x^m, \Delta y^m; \Delta x^n, \Delta y^n)$ in the x and y directions for the mth and nth sequences.

All subpixel offsets (m,n) constitute the sets $(\Delta x_{\text{A}}, \Delta y_{\text{A}}; \Delta x_{\text{B}}, \Delta y_{\text{B}})$ between images I_{A} and I_{B}. Under the original image resolution, the fine-level matching results are combined with the subpixel offsets to obtain the final matched point coordinate sets

$$\mathcal{M}_s = \mathcal{M}_f + (\Delta x_{\text{A}}, \Delta y_{\text{A}}; \Delta x_{\text{B}}, \Delta y_{\text{B}}). \tag{17}$$

The subpixel-level optimization not only enhances matching precision to subpixel accuracy but also resolves any remaining one-to-many/many-to-one matches from the previous stage into one-to-one correspondences.

C. Supervision

A multilevel supervised loss function is employed, which includes a coarse-level matching loss $\mathcal{L}_c$, a fine-level matching loss $\mathcal{L}_f$, and a subpixel-level optimization loss $\mathcal{L}_s$. The design of each component is detailed below.

1) Coarse-Level Matching Loss: This loss operates on low-resolution feature maps' confidence matrices $\mathcal{P}_c^{i\to j}$ and $\mathcal{P}_c^{j\to i}$. At this stage, we adopt the focal loss [50], focusing solely on positive matching samples. Given the ground-truth matches $\mathcal{M}_c^{\text{gt}}$, the loss function is defined as

$$\begin{aligned}\mathcal{L}_c = -\frac{\alpha}{|\mathcal{M}_c^{\text{gt}}|}\sum_{(i,j)\in\mathcal{M}_c^{\text{gt}}}&\left[\left(1-\mathcal{P}_c^{i\to j}\right)^{\gamma}\log\mathcal{P}_c^{i\to j}\right. \\ &\left.+\left(1-\mathcal{P}_c^{j\to i}\right)^{\gamma}\log\mathcal{P}_c^{j\to i}\right]\end{aligned} \tag{18}$$

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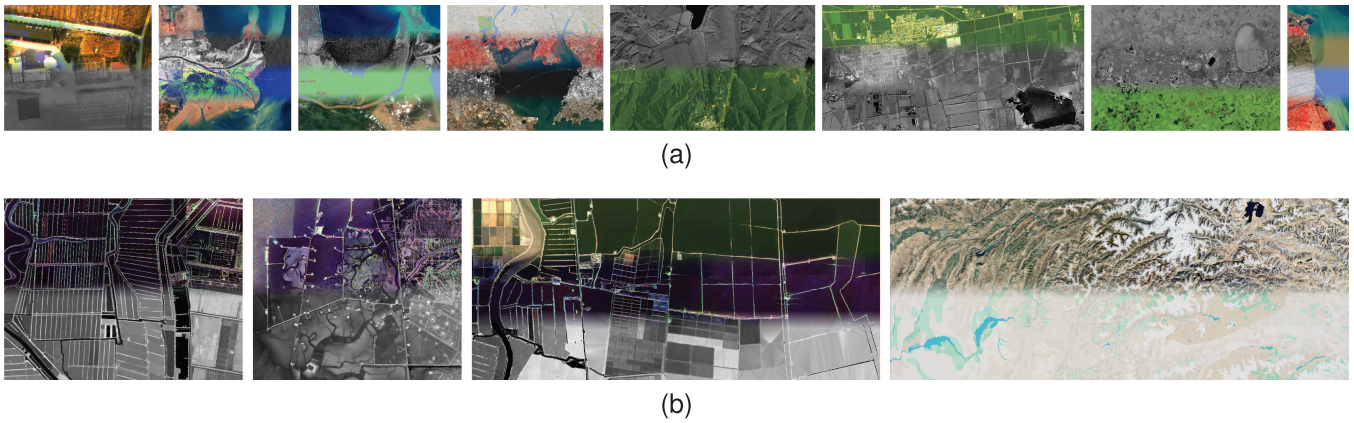


Fig. 3. Overview of each scene in the introduced data, including 8 training scenes and 4 test scenes. (a) Training data: Scenes 1 ~ 8 are arranged from left to right. (b) Testing data: Scenes 1 ~ 4 are arranged from left to right.

where $|\cdot|$ is the operation of calculating the size of a set, α denotes the balancing factor, and γ represents the focusing parameter for hard samples, with $\alpha = 0.25$ and $\gamma = 2$.

2) Fine-Level Matching Loss: For high-resolution window-level matching, a symmetric Focal Loss supervises the confidence matrix $\mathcal{P}_f = \{\mathcal{P}_f^{(i,j)} | (i,j) \in \mathcal{M}_c\}$. Given the ground truth matches $\mathcal{M}_f^{\text{gt}}$, we define positive and negative sample sets as $\mathcal{M}_f^{\text{pos}} = \{(m,n) | (m,n) \in \mathcal{M}_f, (m,n) \in \mathcal{M}_f^{\text{gt}}\}$ and $\mathcal{M}_f^{\text{neg}} = \{(m,n) | (m,n) \in \mathcal{M}_f, (m,n) \notin \mathcal{M}_f^{\text{gt}}\}$, respectively. The loss function is formulated as

$$\mathcal{L}_f = -\alpha \left[\frac{1}{\left|\mathcal{M}_f^{\text{pos}}\right|} \sum_{(m,n)\in\mathcal{M}_f^{\text{pos}}} \left(1 - \mathcal{P}_f\right)^{\gamma} \log \mathcal{P}_f + \frac{1}{\left|\mathcal{M}_f^{\text{neg}}\right|} \sum_{(m,n)\in\mathcal{M}_f^{\text{neg}}} \mathcal{P}_f{}^{\gamma} \log\left(1 - \mathcal{P}_f\right) \right] \tag{19}$$

where the values of α and γ remain consistent with those used in the coarse-level matching loss.

3) Subpixel-Level Optimization Loss: To enhance the localization accuracy of matching points, we introduce a reprojection error-based supervision term. Given a pair of matching points $\{p_{\text{A}}^{(k)}, p_{\text{B}}^{(k)}\} \in \mathcal{M}_s$, where $k = 1, 2, \ldots, |\mathcal{M}_s|$, and the ground-truth homography matrix from image I_{A} to I_{B} as $T_{\text{A}\to\text{B}}$, the geometric consistency constraint is computed as

$$\mathcal{L}_s = \frac{1}{|\mathcal{S}|} \sum_{k\in\mathcal{S}} \left\| T_{\text{A}\to\text{B}} \circ p_{\text{A}}^{(k)} - p_{\text{B}}^{(k)} \right\|_2 \tag{20}$$

where $\circ$ denotes the homography transformation, and $\mathcal{S} = \{k | \|T_{\text{A}\to\text{B}} \circ p_{\text{A}}^{(k)} - p_{\text{B}}^{(k)}\| \le 5\}$ represents a filtered subset of reliable matches. Specifically, a match $\{p_{\text{A}}^{(k)}, p_{\text{B}}^{(k)}\}$ is considered valid only if the Euclidean distance between the transformed position of $p_{\text{A}}^{(k)}$ and $p_{\text{B}}^{(k)}$ is within 5 pixels. This design ensures that the network learns subpixel-level offsets exclusively from geometrically consistent matches.

4) Total Loss: The final loss combines the three components

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_c + \lambda_2 \mathcal{L}_f + \lambda_3 \mathcal{L}_s \tag{21}$$

where the weighting coefficients for each loss term are set to $\lambda_1 = 0.5$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.1$, respectively.

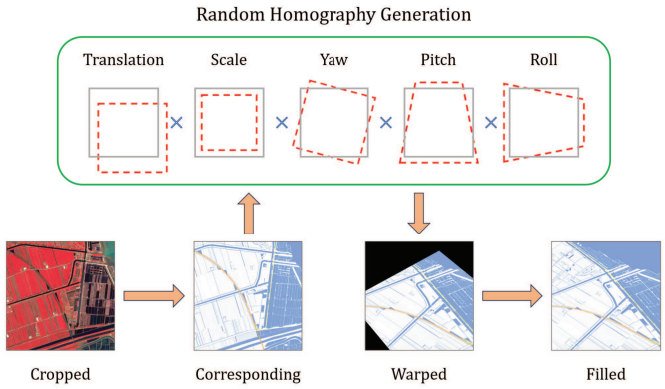


Fig. 4. Proposed data sample generation strategy.

IV. Experiments

A. Datasets

To train and evaluate the robustness and generalization of the proposed model in cross-modal image matching, we collect a diverse dataset comprising 12 scenes that represent as many modalities as possible in remote sensing images. The data visualization is shown in Fig. 3, with details of each scene listed in Table I. Data from eight scenes are used for model training, while data from the remaining four scenes are reserved for testing.

For training and testing sample generation, fixed-size patches are cropped from corresponding large images of two modalities in each scene, maintaining identical spatial positions. In each cropped pair, one image undergoes several random homography transformations. The proposed data sample generation strategy is illustrated in Fig. 4. In addition to translation and scaling, we apply central rotation and symmetric perspective distortions along the x and y directions to simulate yaw, pitch, and roll angular changes. These transformations are combined to generate a random homography matrix. Each sample consists of a transformed patch from one modality, untransformed patches from both modalities, and the generated homography matrix as the label. To simulate realistic remote sensing images captured from

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TABLE I
DETAILS OF EACH SCENE IN THE INTRODUCED TRAINING AND TESTING DATA

Type	Data	Resolution	Sample Size	Modalities				
				Visible	Infrared	SAR	LiDAR	Map
Training	Scene 1	1300×1106	6872	✓	✓			
	Scene 2	4022×3552		✓		✓		
	Scene 3	2453×1972		✓		✓		✓
	Scene 4	4230×3220		✓		✓		✓
	Scene 5	5757×2861		✓		✓		
	Scene 6	5544×3009		✓		✓	✓	
	Scene 7	4983×3246		✓		✓	✓	
	Scene 8	1755×3526		✓				✓
Testing	Scene 1	4797×3643	90			✓	✓	
	Scene 2	4525×3907	90			✓	✓	
	Scene 3	5150×2222	120	✓		✓	✓	
	Scene 4	1900×750	100	✓				✓

varying viewpoints, the transformation is applied to the larger area surrounding the cropped patch, in order to avoid the blank border regions.

Specifically, for training data across all scenes and testing data from Scene 4, patches are cropped using a 512×512 window. For testing data from Scenes $1 \sim 3$, which contain excessive repetitive textures or textureless regions, we first crop larger regions with a 2048×2048 window, then resize them to 512×512. All cropped patches undergo 10 random homography transformations. Finally, generated samples are filtered to remove featureless regions, such as pure ocean areas. The testing dataset comprises 400 samples covering four modalities: visible light, SAR, LiDAR, and map.

Additionally, we evaluate model performance on the publicly available MRSI dataset [9], [51], which includes six modalities (optical, infrared, SAR, depth, map, and nighttime images) with diverse representations, making it highly suitable for comprehensive testing of cross-modal matching capabilities.

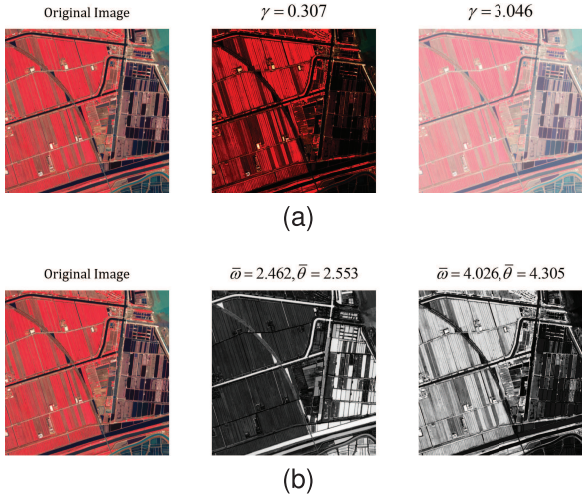


Fig. 5. Demonstration of the two data augmentation strategies employed under varying parameters. (a) Gamma mapping. (b) Thermal mapping.

B. Implementation Details

The model is trained on a single NVIDIA GeForce RTX 4090D (24GB) GPU, with an input image size of 512×512, and a batch size set to 2. To facilitate loss convergence and promote learning of shared features across modalities, a three-stage training strategy is adopted. In Stage 1, the model is trained for 10 epochs on single-modal image pairs with geometric transformations, enabling it to acquire basic invariant feature perception capabilities. In Stage 2, the training runs for 30 epochs on single-modal image pairs with geometric transformations. During the first two stages, random gamma mapping is applied to both input images to simulate illumination variations as a form of data augmentation. In Stage 3, one input image undergoes random thermal mapping instead of gamma mapping, and training continues until convergence. The AdamW optimizer is used with a weight decay of 0.1. The initial learning rates for Stages 1 and 2 are set to 2.5×10^{-4}, including a 5-epoch warm-up. For Stage 3, the initial learning rate is 6.25×10^{-5}.

1) Gamma Mapping: Since the training data does not contain any scenes with the day–night modality pair, we introduce random gamma mapping to simulate illumination variations between day and night. This operation is applied throughout all training stages. For an input image $I \in [0, 255]$, the gamma-mapped output is

$$I_\gamma = \left(\frac{I}{255}\right)^{\frac{1}{\gamma}} \cdot 255 \tag{22}$$

where γ is a nonlinear correction parameter. It is randomly selected from $[0.3, 0.6]$, 1, or $[2, 4]$, each with an equal probability of $(1/3)$. When $\gamma = 1$, no transformation is applied. For the other two intervals, brightness is either reduced or enhanced, as shown in Fig. 5(a).

2) Thermal Mapping: Inspired by [13] and [52], we introduce random thermal mapping to simulate pseudo-infrared images, compensating for the limited availability of infrared data during training. This operation, executed only in Stage 3, employs a cosine function. For input image $I \in [0, 255]$, the thermal mapping can be expressed as

$$I_{\text{temp}} = \cos\left(\bar{\omega} \cdot \left(\frac{I}{255} - 0.5\right) + \bar{\theta}\right) \tag{23}$$

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$$I_{\text{thermal}} = \frac{I_{\text{temp}} - \min\left(I_{\text{temp}}\right)}{\max\left(I_{\text{temp}}\right) - \min\left(I_{\text{temp}}\right)} \cdot 255 \tag{24}$$

where I_{temp} is the intermediate image before normalization, I_{thermal} is the final pseudo-infrared output, $\bar{\omega} = (2\pi/3) + |\alpha_1| \cdot (\pi/2)$, $\bar{\theta} = (\pi/2) + \alpha_2 \cdot (\pi/2)$, and random variables α_1 and α_2 follow a standard normal distribution. Examples of thermal mapping are shown in Fig. 5.

C. Evaluation Protocol

SwinMatcher is compared with 13 baseline and state-of-the-art methods: RIFT [9], D2-Net [32], SuperPoint [28] +LightGlue [34] (SP+LG), LoFTR [35], ASpanFormer [36], MatchFormer [37], TopicFM [53], Efficient LoFTR [39] (ELoFTR), and XoFTR [13], as well as four methods fine-tuned with MINIMA [54], including LightGlue ($\text{MINIMA}_{\text{LG}}$), LoFTR ($\text{MINIMA}_{\text{LoFTR}}$), Efficient LoFTR ($\text{MINIMA}_{\text{ELoFTR}}$), and XoFTR ($\text{MINIMA}_{\text{XoFTR}}$). RIFT is a representative traditional cross-modal matching method in remote sensing, while others are deep learning-based. MINIMA is a training framework that fine-tunes image matching models using richly synthesized cross-modal data. RIFT, D2-Net, SP+LG, and SP+$\text{MINIMA}_{\text{LG}}$ are detector-based methods, with the first two using NN matching, and the others are detector-free semidense matchers. TopicFM-fast, an enhanced version of TopicFM, is adopted for its stronger cross-modal robustness. For SwinMatcher, coarse and fine matching thresholds are set to $\theta_c = 0.3$ and $\theta_f = 0.1$, respectively. All baselines use default parameters.

Four metrics are used to evaluate matching performance: the number of correct matches (NCMs), precision (Pre), success rate (SR), and root mean square error (RMSE). Following [27], matches are directly filtered using labels without RANSAC-like [55] outlier removal to avoid eliminating valid ones. The metrics are defined as follows.

1) *NCM:* For a match pair $\{p, q\}$, if the Euclidean distance between the transformed position p' (via label) and q is ≤ 5 pixels, the match is considered correct.
2) *Pre:* Precision is the ratio of correct matches to the total matches output by each method.
3) *SR:* An image pair is deemed successful if it contains ≥ 20 correct matches. SR is the ratio of successful pairs to the total pairs.
4) *RMSE:* For successful pairs, RMSE is computed as

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{p}_i' - \tilde{q}_i\right\|_2^2} \tag{25}$$

where $(\tilde{p}_i, \tilde{q}_i)$ are correct matches, $\tilde{p}_i'$ is the transformed position of $\tilde{p}_i$, and N is the NCMs. For failed pairs, RMSE is set to 10. During experiments, SR and the average NCM, Pre, and RMSE of each method are computed separately for each modality pair.

D. Experiments on MRSI Dataset

The proposed method is first evaluated on the original MRSI dataset. This public dataset includes ten samples for each of six modality pairs: optical–optical, optical–infrared, optical–SAR, optical–depth, optical–map, and day–night. Since different methods may have specific requirements for input image sizes (e.g., divisibility by 8, 16, or 32), to ensure fairness, input images are scaled to a uniform size 512×512 compatible with all methods, and the coordinates of matched points are scaled back to the original image size for label calculation.

Fig. 6 provides a visual comparison of the performance metrics across all the methods on the original MRSI dataset. The results demonstrate that SwinMatcher achieves the highest NCM and an SR of 100.0% across all modality pairs. It also maintains high Pre and consistently good RMSE. However, other methods, particularly XoFTR, exhibit competitive performance, with the lowest RMSE in most modalities. Considering that the original MRSI dataset has limitations, such as a small sample size with only ten samples per modality pair, the statistical significance of the experimental results is limited. Additionally, most of the samples in each modality pair only involve simple translation transformations, with only two samples exhibiting noticeable rotation or scale changes. As a result, the overall matching task is relatively easy, and most methods achieve high SR across all modality pairs. This makes it difficult to effectively assess the model's robustness to complex cross-modal geometric distortions and to demonstrate the advantages of SwinMatcher.

To address this, we expand the dataset by precisely aligning image pairs using ground-truth homography matrices, resizing them to 512×512, and applying ten random homography transformations to one modality in each pair. This operation expands the dataset to 600 samples, allowing for a more thorough and rigorous evaluation. The quantitative results on the expanded MRSI dataset are presented in Tables II and III, with a visual comparison of performance metrics illustrated in Fig. 7 and matching examples shown in Fig. 8.

The traditional method RIFT performs poorly across all metrics, underscoring the limitations of handcrafted features (e.g., phase congruency) in handling cross-modal geometric distortions. D2-Net produces low average NCM and Pre due to its reliance on CNN-based local descriptors, which cannot capture global semantic relationships. SP+LG achieves competitive Pre in optical–optical, optical–depth, and day–night but suffers from severe performance degradation in optical–SAR, indicating sensitivity to modality differences. Detector-free methods LoFTR, MatchFormer, TopicFM, and ELoFTR perform well in optical–optical but show limited cross-modal robustness, with low NCM and Pre in other modalities. For instance, ELoFTR achieves the lowest RMSE (2.159) in optical–optical due to extensive training on optical data but fails to generalize to other modalities. The generalization ability of ASpanFormer is particularly limited. It shows lower SR and higher average RMSE in most modalities beyond optical–optical and optical–depth cases. Its dynamic adjustment of the local attention range through probabilistic modeling mainly relies on flow map regression for adaptive span. It is hypothesized that in complex cross-modal scenes, if deformations exceed the local adjustment range predicted by the flow map, matching failures may occur. XoFTR achieves low average RMSE in optical–infrared and

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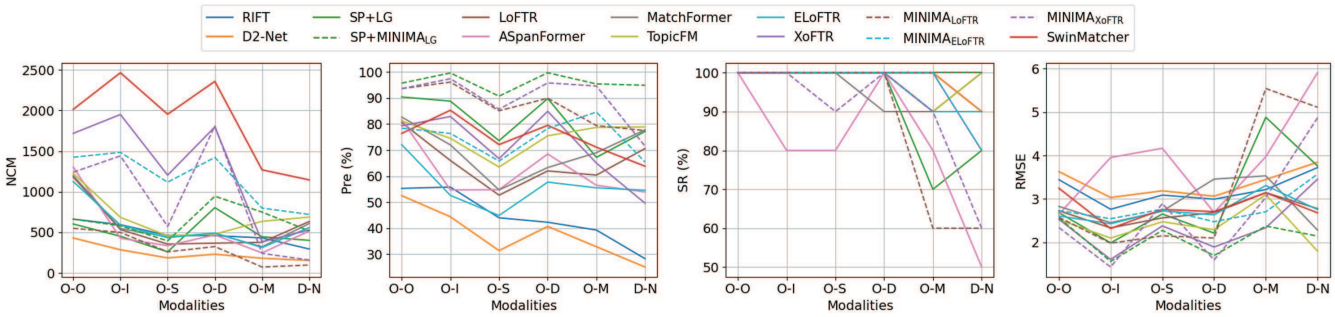


Fig. 6. Visual comparison of performance metrics across all the methods on the original MRSI dataset, covering six modality pairs: O-O (optical–optical), O-I (optical–infrared), O-S (optical–SAR), O-D (optical–depth), O-M (optical–map), and D-N (day–night).

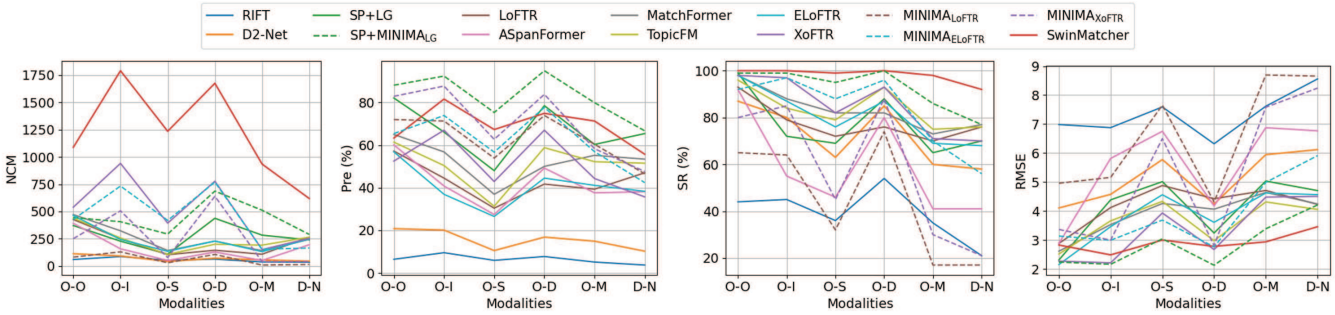


Fig. 7. Visual comparison of performance metrics across all the methods on the expanded MRSI dataset, covering six modality pairs: O-O (optical–optical), O-I (optical–infrared), O-S (optical–SAR), O-D (optical–depth), O-M (optical–map), and D-N (day–night).

TABLE II
QUANTITATIVE RESULTS OF EACH METHOD ON THE EXPANDED MRSI DATASET (PART I)

Categories	Methods	Modality Pairs											
		Optical-Optical				Optical-Infrared				Optical-SAR			
		NCM	Pre (%)	SR (%)	RMSE	NCM	Pre (%)	SR (%)	RMSE	NCM	Pre (%)	SR (%)	RMSE
Sparse	RIFT	60.4	6.4	44.0	6.984	87.7	9.5	45.0	6.878	53.8	5.9	36.0	7.591
	D2-Net	113.8	20.8	87.0	4.109	93.2	20.1	80.0	4.578	42.1	10.5	63.0	5.777
	SP+LG	371.8	82.1	99.0	2.316	228.3	66.1	72.0	4.389	109.7	48.0	69.0	5.016
	SP+MINIMA$_{LG}$	444.7	**88.1**	99.0	2.245	406.5	**92.4**	99.0	**2.168**	293.7	**75.3**	95.0	3.047
Semi-Dense	LoFTR	426.7	57.2	93.0	2.886	251.8	44.5	79.0	4.123	104.7	30.4	72.0	4.883
	ASpanFormer	391.8	60.1	92.0	2.888	158.4	40.8	55.0	5.816	51.6	27.4	46.0	6.753
	MatchFormer	472.0	65.0	98.0	2.619	323.3	56.8	88.0	3.538	142.9	36.9	82.0	4.263
	TopicFM	439.6	61.4	96.0	2.522	257.7	50.4	84.0	3.666	106.8	31.3	79.0	4.337
	ELoFTR	463.5	56.8	98.0	**2.159**	244.0	36.9	87.0	3.467	135.4	26.5	76.0	4.564
	XoFTR	539.3	52.5	98.0	2.278	940.8	67.0	97.0	2.218	393.8	42.9	82.0	3.935
	MINIMA$_{LoFTR}$	82.4	72.0	65.0	4.962	130.8	71.3	64.0	5.163	30.9	53.9	32.0	7.654
	MINIMA$_{ELoFTR}$	427.5	65.4	92.0	3.142	733.1	73.9	97.0	2.996	418.6	56.7	88.0	3.692
	MINIMA$_{XoFTR}$	251.2	82.9	80.0	3.368	506.3	87.7	85.0	3.002	78.4	62.7	45.0	6.500
	SwinMatcher (Ours)	**1087.2**	63.3	**100.0**	2.831	**1788.8**	81.6	**100.0**	2.493	**1234.7**	67.3	**99.0**	**3.003**

optical–depth, which can be attributed to its training on optical and infrared data, as well as the use of effective data augmentation strategies. However, its performance in optical–map and day–night scenes is suboptimal, indicating limited consideration of modality diversity and suggesting that the model's cross-modal generalization capability still requires improvement. For methods fine-tuned with MINIMA, SP+MINIMA$_{LG}$ achieves the highest average Pre across all modality pairs and the lowest average RMSE in optical–infrared (2.168) and optical–depth (2.135), demonstrating the advancement of this training framework. MINIMA$_{ELoFTR}$ also achieves an improvement in cross-modal matching performance, whereas MINIMA$_{LoFTR}$ and MINIMA$_{XoFTR}$ suffer significant degradation, presumably due to overfitting to the training data.

SwinMatcher demonstrates superior performance on the expanded MRSI dataset. It achieves approximately double the average NCM of the second-best method across all modality pairs. The SR approaches or reaches 100% in all modality pairs except day–night, validating robustness to complex geometric and radiometric variations. Notably, although the SR in day–night reaches the highest value of 92%, it is not flawless. The main reason is that the dataset contains a wide range of modalities, each with multiple appearance variations, making it challenging for the training data to cover all possible types, especially for the day–night modality pair. In particular, when certain unseen appearance variations coincide with severe geometric transformations, the model may encounter failure cases. Moreover, SwinMatcher maintains stable Pre and

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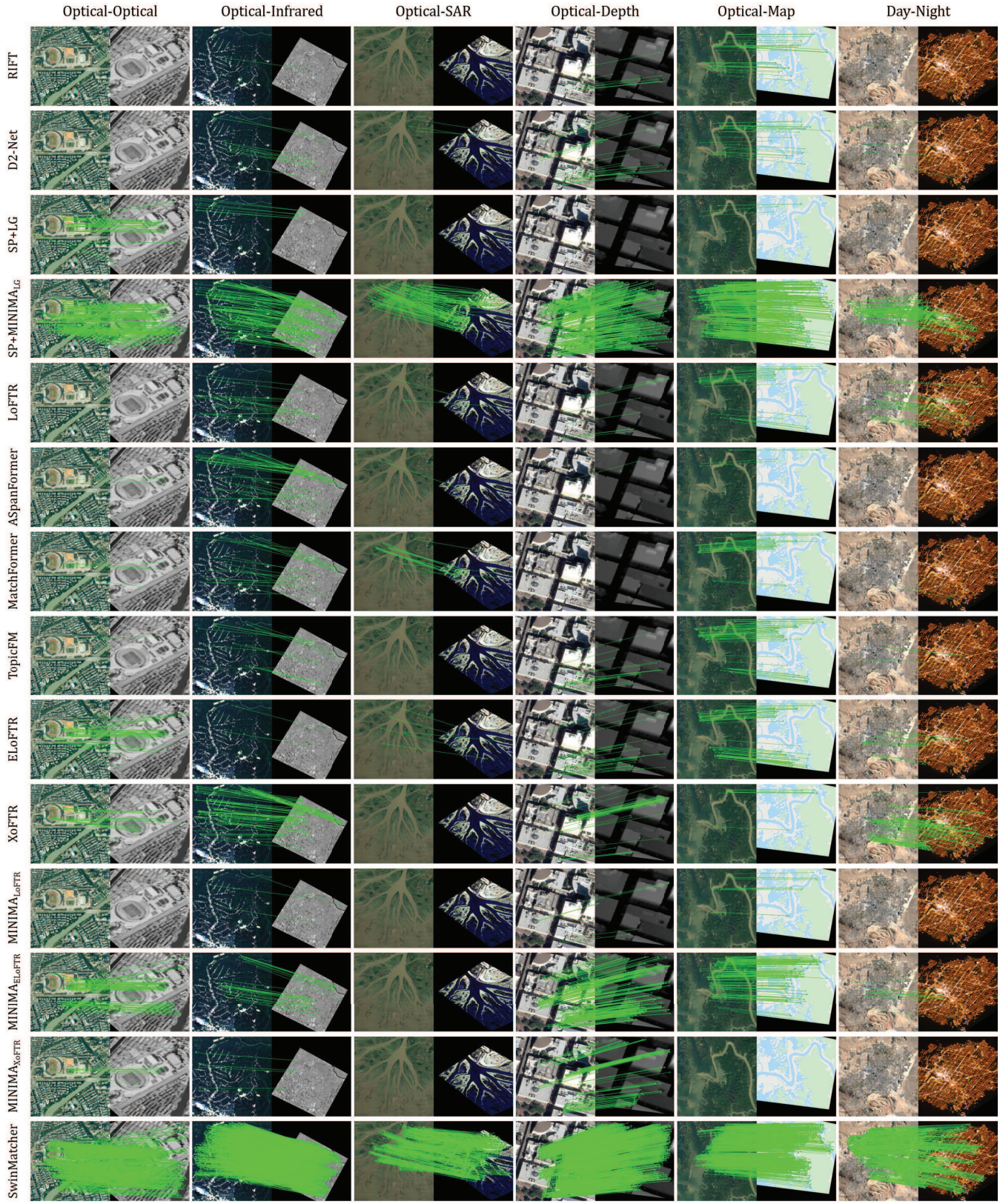


Fig. 8. Qualitative results of each method on the expanded MRSI dataset. For each modality pair, one matching example is selected, connecting only the correctly matched points between images with green lines.

RMSE, achieving the lowest RMSE in optical–SAR (3.003), optical–map (2.941), and day–night (3.463). These results highlight its ability to perform high-precision and reliable matching in challenging cross-modal scenes.

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TABLE III
QUANTITATIVE RESULTS OF EACH METHOD ON THE EXPANDED MRSI DATASET (PART II)

Categories	Methods	Modality Pairs											
		Optical-Depth				Optical-Map				Day-Night			
		NCM	Pre (%)	SR (%)	RMSE	NCM	Pre (%)	SR (%)	RMSE	NCM	Pre (%)	SR (%)	RMSE
Sparse	RIFT	63.9	7.7	54.0	6.322	38.3	5.1	35.0	7.616	36.8	3.7	21.0	8.557
	D2-Net	72.6	16.8	85.0	4.246	56.6	14.9	60.0	5.947	46.2	10.2	58.0	6.120
	SP+LG	438.0	78.5	88.0	3.246	283.5	60.3	65.0	5.037	240.0	65.4	70.0	4.708
	SP+MINIMA$_{\mathrm{LG}}$	686.9	**94.9**	**100.0**	**2.135**	511.7	**79.9**	86.0	3.392	290.2	**66.7**	77.0	4.213
Semi-Dense	LoFTR	145.3	41.7	76.0	4.429	109.1	39.3	70.0	4.713	264.6	47.0	76.0	4.233
	ASpanFormer	129.4	49.2	80.0	4.180	51.8	37.6	41.0	6.876	195.9	38.1	41.0	6.768
	MatchFormer	229.1	50.0	82.0	4.067	141.8	55.2	73.0	4.645	244.4	53.4	77.0	4.248
	TopicFM	200.4	58.8	93.0	2.981	192.1	52.2	75.0	4.318	265.2	51.5	76.0	4.059
	ELoFTR	229.6	44.5	87.0	3.615	132.5	41.1	69.0	4.626	248.2	38.3	68.0	4.568
	XoFTR	777.5	67.0	93.0	2.677	151.7	44.3	71.0	4.484	254.3	35.7	70.0	4.508
	MINIMA$_{\mathrm{LoFTR}}$	107.2	73.9	74.0	4.318	9.9	60.4	17.0	8.694	15.5	46.5	17.0	8.654
	MINIMA$_{\mathrm{ELoFTR}}$	769.4	77.7	96.0	2.769	154.1	56.9	70.0	4.990	165.9	42.4	56.0	5.913
	MINIMA$_{\mathrm{XoFTR}}$	639.6	83.8	88.0	2.812	41.4	58.3	30.0	7.582	41.6	47.6	21.0	8.240
	SwinMatcher (Ours)	**1674.2**	74.9	**100.0**	2.785	**934.2**	71.3	**98.0**	**2.941**	**618.8**	55.7	**92.0**	**3.463**

E. Experiments on Proposed Dataset

To further evaluate the cross-modal generalization of the image matching model, we conduct experimental validation on the proposed testing dataset which comprises 400 samples across four scenes: Scenes 1 and 2 each include 90 SAR-LiDAR samples, Scene 3 contains 50 visible-LiDAR and 70 visible-SAR samples, and Scene 4 includes 100 visible-map samples. All images are uniformly resized to 512×512, with random homography transformations applied between each image pair. The quantitative results on the proposed dataset are summarized in Tables IV and V, with a visual comparison of performance metrics illustrated in Fig. 9, and matching examples for each modality pair shown in Fig. 10.

Experimental results indicate that the traditional method, RIFT, still struggles with complex perspective distortions in cross-modal scenes, consistently ranking last in all metrics. The simplistic model architecture of D2-Net hinders the learning of robust matching features, resulting in poor performance across all modalities. SP+LG attains high average Pre across all scenes and low average RMSE in SAR-LiDAR (Scene 2) and visible-SAR (Scene 3), demonstrating high confidence and precision in matched feature points and partial cross-modal capability. However, its performance drops significantly in visible-LiDAR (Scene 3) and visible-map (Scene 4), highlighting its instability in cross-modal settings. LoFTR, MatchFormer, TopicFM, and ELoFTR, under the same detector-free framework, exhibit mediocre and comparable performance, showing limited robustness for cross-modal image matching. ASpanFormer underperforms these four methods, suggesting that its dynamic span prediction mechanism may not adapt well to cross-modal complexities. XoFTR delivers competitive performance in SAR-LiDAR (Scenes 1 and 2) and visible-SAR (Scene 3), but its performance deteriorates when applied to other modalities, revealing limitations in cross-modal generalization. For methods fine-tuned with MINIMA, SP+MINIMA$_{\mathrm{LG}}$ achieves the highest average Pre in all scenes and the lowest average RMSE in SAR-LiDAR (Scene 1: 2.143, Scene 2: 2.084) and visible-SAR (Scene 3: 1.810), exhibiting strong generalization. Still, due to potential overfitting, MINIMA$_{\mathrm{LoFTR}}$ and MINIMA$_{\mathrm{XoFTR}}$ exhibit inferior matching generalization relative to their original training performance.

SwinMatcher demonstrates comprehensive superiority on the new dataset. In all four scenes, its average NCM significantly surpasses other methods, with its average Pre remaining competitive. Moreover, its SR approaches or reaches 100% in all scenes, with average RMSE consistently maintained at low levels. Notably, SwinMatcher achieves the lowest RMSE of 2.551, and 3.110 in visible-LiDAR (Scene 3), and visible-map (Scene 4), respectively. The high matching accuracy, robustness, and generalization across diverse scenes and modalities validate the effectiveness of our training data, strategies, and the model's practical utility.

F. Ablation Study

To further validate the effectiveness of our training strategies and the rationality of key module designs in SwinMatcher, ablation experiments are conducted by controlling a single variable in the following five configurations.

1) Removing gamma mapping in training stage 3) while retaining only thermal mapping for data augmentation.
2) Omitting thermal mapping entirely, i.e., limiting training to two stages.
3) Reverting to the original Swin Transformer Blocks, which exclude window/SWCA mechanisms.
4) Adopting only one-to-one correspondence during the coarse matching stage, discarding one-to-many/many-to-one relationships.
5) Removing the proposed subpixel regression convolution block, i.e., performing no subpixel-level optimization for matches. We compare the average NCM of these five configurations against the full model on the expanded MRSI dataset.

As shown in Table VI, gamma mapping and thermal mapping complement each other, and neither should be omitted. The introduced window/SWCA mechanisms critically enhance the model's matching robustness. Retaining one-to-many/many-to-one correspondences during coarse matching significantly improves the final NCMs. The proposed subpixel regression convolution block also contributes moderately to

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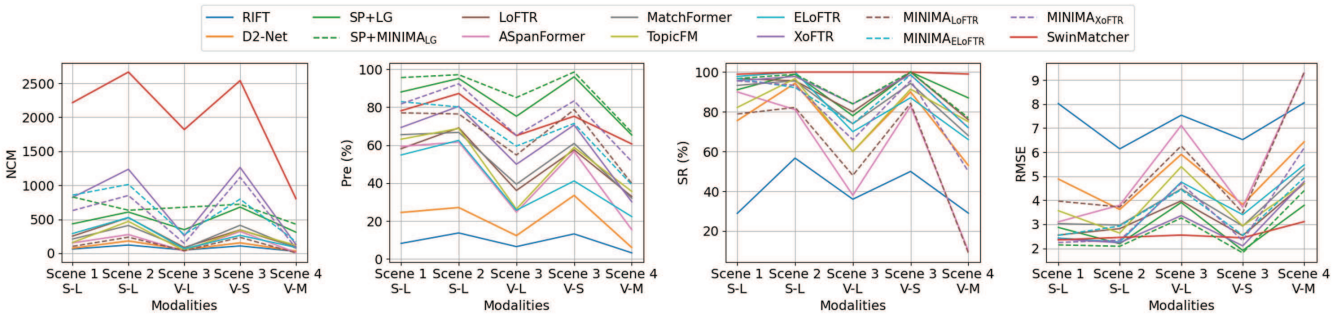


Fig. 9. Visual comparison of performance metrics across all the methods on the proposed dataset, covering four modality pairs: S-L (SAR-LiDAR), V-L (visible-LiDAR), V-S (visible-SAR), and V-M (visible-map).

TABLE IV

QUANTITATIVE RESULTS OF EACH METHOD ON THE PROPOSED DATASET (PART I)

Categories	Methods	Modality Pairs							
		Scene 1: SAR-LiDAR				Scene 2: SAR-LiDAR			
		NCM	Pre (%)	SR (%)	RMSE	NCM	Pre (%)	SR (%)	RMSE
Sparse	RIFT	62.9	8.2	28.9	8.022	121.6	13.7	56.7	6.138
	D2-Net	75.0	24.5	75.6	4.884	182.1	27.1	94.4	3.616
	SP+LG	433.4	88.0	91.1	2.870	606.9	95.1	98.9	2.191
	SP+MINIMA$_{LG}$	828.7	**95.6**	96.7	**2.143**	633.6	**97.1**	98.9	**2.084**
Semi-Dense	LoFTR	253.1	58.1	96.7	2.553	524.9	69.1	95.6	2.818
	ASpanFormer	153.1	59.3	90.0	3.101	275.4	61.6	81.1	3.794
	MatchFormer	210.0	65.5	93.3	3.028	409.1	66.7	95.6	2.967
	TopicFM	159.1	63.1	82.2	3.566	469.7	68.8	96.7	2.632
	ELoFTR	290.8	54.8	97.8	2.426	526.6	62.5	**100.0**	2.266
	XoFTR	822.6	69.3	95.6	2.382	1236.4	80.4	97.8	2.235
	MINIMA$_{LoFTR}$	101.8	77.0	78.9	3.963	236.9	76.4	82.2	3.721
	MINIMA$_{ELoFTR}$	856.4	82.9	96.7	2.541	1012.8	80.1	92.2	2.927
	MINIMA$_{XoFTR}$	628.7	81.7	95.6	2.244	850.3	92.3	93.3	2.336
	SwinMatcher (Ours)	**2216.8**	78.1	**98.9**	2.352	**2666.2**	87.2	100.0	2.453

TABLE V

QUANTITATIVE RESULTS OF EACH METHOD ON THE PROPOSED DATASET (PART II)

Categories	Methods	Modality Pairs											
		Scene 3: Visible-LiDAR				Scene 3: Visible-SAR				Scene 4: Visible-Map			
		NCM	Pre (%)	SR (%)	RMSE	NCM	Pre (%)	SR (%)	RMSE	NCM	Pre (%)	SR (%)	RMSE
Sparse	RIFT	48.4	6.5	36.0	7.538	107.9	13.2	50.0	6.522	29.3	3.2	29.0	8.056
	D2-Net	53.8	12.3	60.0	5.907	153.9	33.5	90.0	3.825	32.6	6.0	53.0	6.432
	SP+LG	347.1	75.2	78.0	3.907	679.2	96.0	**100.0**	1.910	310.5	65.3	87.0	3.791
	SP+MINIMA$_{LG}$	676.9	**85.1**	84.0	3.259	721.3	**98.5**	**100.0**	**1.810**	427.9	**66.7**	77.0	4.402
Semi-Dense	LoFTR	85.6	36.1	80.0	3.975	344.8	57.5	**100.0**	2.526	94.5	32.9	76.0	4.698
	ASpanFormer	31.3	24.7	38.0	7.114	321.5	56.6	82.9	3.707	11.8	15.4	10.0	9.287
	MatchFormer	74.1	39.4	74.0	4.474	412.1	60.9	94.3	2.926	75.5	32.0	68.0	5.316
	TopicFM	39.5	26.3	60.0	5.397	340.4	58.8	91.4	2.933	119.6	35.6	75.0	4.707
	ELoFTR	66.4	25.7	70.0	4.756	263.7	41.1	87.1	3.399	87.6	22.3	66.0	5.472
	XoFTR	257.3	49.9	84.0	3.358	1263.5	70.6	98.6	2.102	127.5	30.1	72.0	4.779
	MINIMA$_{LoFTR}$	38.3	54.8	48.0	6.260	235.1	78.7	84.3	3.471	6.2	39.8	9.0	9.333
	MINIMA$_{ELoFTR}$	274.3	59.5	74.0	4.440	797.6	71.4	98.6	2.546	170.8	39.1	72.0	4.943
	MINIMA$_{XoFTR}$	146.7	65.1	66.0	4.709	1119.1	83.3	95.7	2.317	75.6	51.3	50.0	6.182
	SwinMatcher (Ours)	**1820.2**	65.0	**100.0**	**2.551**	**2540.2**	75.2	**100.0**	2.437	**802.8**	60.6	**99.0**	**3.110**

matching precision, thereby increasing the count of correct matches. These findings validate the rationality behind our training strategies and key module designs, each contributing positively to the model's performance.

G. Running Time Analysis

Matching efficiency tests are conducted for each method on a Windows system, with an AMD Ryzen 7 5800X 8-Core Processor (3.80 GHz) CPU and an NVIDIA GeForce RTX 3070 (8GB) GPU. The test involves 600 samples from the expanded MRSI dataset, with each input image resized to 512×512. To measure the average matching time per image pair, we exclude the first 100 samples as a warm-up phase and compute results based on the remaining 500 samples. The average matching time for each method per image pair is listed in Table VII, along with the average NCM and SR.

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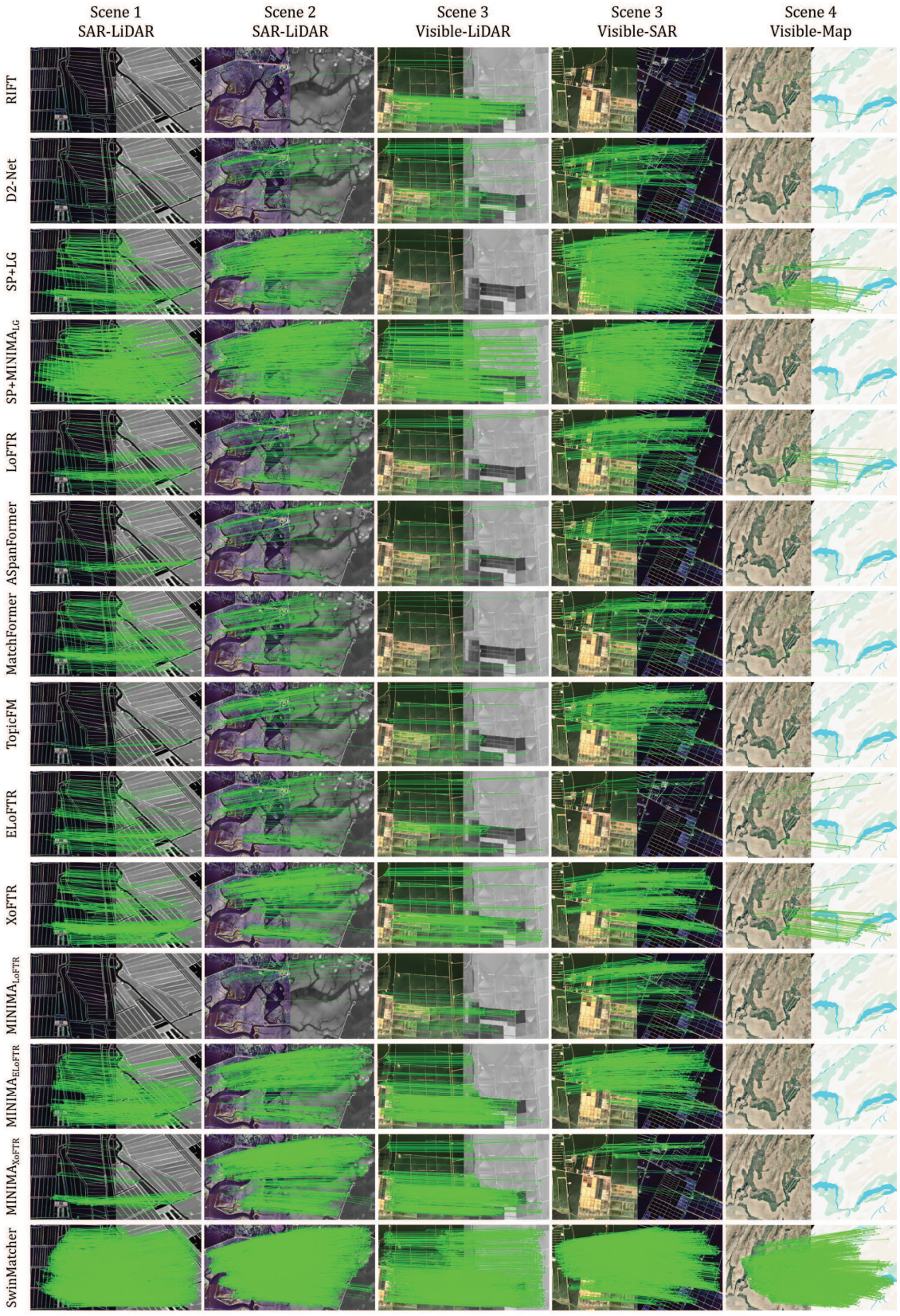


Fig. 10. Qualitative results of each method on the proposed dataset. For each modality pair in each scene, one matching example is selected, connecting only the correctly matched points between images with green lines.

The results show that RIFT, relying on handcrafted features without GPU acceleration, incurs the longest average processing time (3883.1 ms). ELoFTR, employing attention aggregation and other acceleration strategies, records the

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TABLE VI
ABLATION STUDY OF THE IMPACT OF KEY MODULES AND STRATEGIES ON THE AVERAGE NCM ON THE EXPANDED MRSI DATASET

Methods	Average NCM in each modality pair					
	Optical-Optical	Optical-Infrared	Optical-SAR	Optical-Depth	Optical-Map	Day-Night
(1) w/o gamma mapping	1071.5	1775.4	1187.9	1649.6	912.0	591.7
(2) w/o thermal mapping	919.0	1249.8	913.8	1086.0	771.4	448.9
(3) w/o window/shifted-window cross-attention	629.2	1235.1	621.7	858.6	640.3	399.7
(4) w/ only one-to-one coarse matching	614.3	1087.5	721.5	930.5	531.1	348.1
(5) w/o sub-pixel optimization	1086.0	**1788.9**	1233.0	1672.8	933.6	618.6
SwinMatcher (Ours)	**1087.2**	1788.8	**1234.7**	**1674.2**	**934.2**	**618.8**

TABLE VII
COMPARISON OF THE AVERAGE MATCHING TIME OF EACH METHOD ON THE EXPANDED MRSI DATASET

Categories	Methods	NCM	SR (%)	Time (ms)
Sparse	RIFT	56.1	38.4	3883.1
	D2-Net	61.9	68.6	202.3
	SP+LG	258.4	71.6	120.0
Semi-Dense	LoFTR	173.8	74.0	97.4
	ASpanFormer	117.2	52.4	272.2
	MatchFormer	214.8	80.0	246.6
	TopicFM	204.0	80.4	66.6
	ELoFTR	196.4	76.6	**65.4**
	XoFTR	501.4	82.2	97.4
	SwinMatcher (Ours)	**1241.2**	**97.8**	140.2

shortest average processing time (65.4 ms). As a detector-free semidense matcher, SwinMatcher demonstrates competitive efficiency with an average processing time of 140.2 ms, outperforming MatchFormer (a similarly pure Transformer-based method) by more than 100 ms. These results confirm that, while achieving the highest average NCM and SR, SwinMatcher imposes minimal constraints on real-time performance. It also validates that SwinMatcher, through its local–global feature co-design, achieves a balanced trade-off among matching accuracy, robustness, and efficiency across a wide range of cross-modal scenes.

V. CONCLUSION

This article proposes SwinMatcher, an end-to-end cross-modal remote sensing image matching model based on Swin Transformer. By introducing window/SWCA mechanisms, SwinMatcher enables efficient cross-modal feature interaction and multiscale contextual modeling. A coarse-to-fine hierarchical matching architecture further enhances semidense matching precision and robustness. Additionally, we construct a novel cross-modal remote sensing image matching dataset, covering visible light, SAR, LiDAR, and map modalities. This dataset incorporates random homography transformations to simulate real-world geometric distortions, providing a new benchmark for cross-modal image matching tasks. Experimental results demonstrate that SwinMatcher significantly outperforms state-of-the-art methods on both the new and the public MRSI datasets. It exhibits exceptional and stable performance across all metrics in scenes with coupled radiometric differences and geometric distortions, while maintaining practical efficiency. Ablation studies validate the effectiveness of data augmentation strategies, cross-attention mechanisms, and the subpixel regression module. This work offers a reliable solution for multisource remote sensing data applications. The code and proposed dataset will be publicly released to support future research.

REFERENCES

[1] J. Ma, Y. Ma, and C. Li, "Infrared and visible image fusion methods and applications: A survey," *Inf. Fusion*, vol. 45, pp. 153–178, Jan. 2019.

[2] L. Tang, H. Zhang, H. Xu, and J. Ma, "Rethinking the necessity of image fusion in high-level vision tasks: A practical infrared and visible image fusion network based on progressive semantic injection and scene fidelity," *Inf. Fusion*, vol. 99, Nov. 2023, Art. no. 101870.

[3] X. Jiang, G. Li, Y. Liu, X.-P. Zhang, and Y. He, "Change detection in heterogeneous optical and SAR remote sensing images via deep homogeneous feature fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1551–1566, 2020.

[4] H. Zhang, G. Ma, and Y. Zhang, "Intelligent-BCD: A novel knowledge-transfer building change detection framework for high-resolution remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5065–5075, 2022.

[5] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardos, "ORB-SLAM: A versatile and accurate monocular SLAM system," *IEEE Trans. Robot.*, vol. 31, no. 5, pp. 1147–1163, Oct. 2015.

[6] S. Haigang, C. Liu, G. Zhe, J. Zhengjie, and X. Chuan, "Overview of multi-modal remote sensing image matching methods," *Acta Geodaetica et Cartographica Sinica*, vol. 51, no. 9, p. 1848, 2022.

[7] H. Xu, J. Yuan, and J. Ma, "MURF: Mutually reinforcing multi-modal image registration and fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12148–12166, Oct. 2023.

[8] Y. Li et al., "SM3Det: A unified model for multi-modal remote sensing object detection," 2024, *arXiv:2412.20665*.

[9] J. Li, Q. Hu, and M. Ai, "RIFT: Multi-modal image matching based on radiation-variation insensitive feature transform," *IEEE Trans. Image Process.*, vol. 29, pp. 3296–3310, 2020.

[10] J. Li, W. Xu, P. Shi, Y. Zhang, and Q. Hu, "LNIFT: Locally normalized image for rotation invariant multimodal feature matching," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5621314.

[11] Z. Fan, Y. Pi, M. Wang, Y. Kang, and K. Tan, "GLS-MIFT: A modality invariant feature transform with global-to-local searching," *Inf. Fusion*, vol. 105, May 2024, Art. no. 102252.

[12] Y. Deng and J. Ma, "ReDFeat: Recoupling detection and description for multimodal feature learning," *IEEE Trans. Image Process.*, vol. 32, pp. 591–602, 2023.

[13] Ö. TuzcuoĞlu, A. Köksal, B. Sofu, S. Kalkan, and A. A. Alatan, "XoFTR: Cross-modal feature matching transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 4275–4286.

[14] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[15] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 205–218.

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[16] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.

[17] J. P. Lewis, "Fast normalized cross-correlation," *Vis. Interface*, vol. 10, no. 1, pp. 120–123, 1995.

[18] Y. Wu, W. Ma, M. Gong, L. Su, and L. Jiao, "A novel point-matching algorithm based on fast sample consensus for image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 43–47, Jan. 2015.

[19] S. Suri and P. Reinartz, "Mutual-information-based registration of TerraSAR-X and ikonos imagery in urban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 2, pp. 939–949, Feb. 2010.

[20] J. Liang, X. Liu, K. Huang, X. Li, D. Wang, and X. Wang, "Automatic registration of multisensor images using an integrated spatial and mutual information (SMI) metric," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 603–615, Jan. 2014.

[21] J. Öfverstedt, J. Lindblad, and N. Sladoje, "Fast computation of mutual information in the frequency domain with applications to global multimodal image alignment," *Pattern Recognit. Lett.*, vol. 159, pp. 196–203, Jul. 2022.

[22] Y. Ye, J. Shan, L. Bruzzone, and L. Shen, "Robust registration of multimodal remote sensing images based on structural similarity," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2941–2958, May 2017.

[23] Y. Ye, L. Bruzzone, J. Shan, F. Bovolo, and Q. Zhu, "Fast and robust matching for multimodal remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9059–9070, Nov. 2019.

[24] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[25] H. Bay, A. Ess, T. Tuytelaars, and L. Van Gool, "Speeded-up robust features (SURF)," *Comput. Vis. Image Understand.*, vol. 110, no. 3, pp. 346–359, Jun. 2008.

[26] C. Gao, W. Li, R. Tao, and Q. Du, "MS-HLMO: Multiscale histogram of local main orientation for remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626714, doi: 10.1109/TGRS.2022.3193109.

[27] J. Li, Q. Hu, and Y. Zhang, "Multimodal image matching: A scale-invariant algorithm and an open dataset," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 77–88, Oct. 2023.

[28] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 224–236.

[29] Z. Luo et al., "GeoDesc: Learning local descriptors by integrating geometry constraints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 168–183.

[30] Z. Luo et al., "ContextDesc: Local descriptor augmentation with cross-modality context," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2527–2536.

[31] J. Revaud, C. R. de Souza, M. Humenberger, and P. Weinzaepfel, "R2D2: Reliable and repeatable detector and descriptor," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 12405–12415.

[32] M. Dusmanu et al., "D2-Net: A trainable CNN for joint description and detection of local features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8092–8101.

[33] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4938–4947.

[34] P. Lindenberger, P. E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17627–17638.

[35] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8922–8931.

[36] H. Chen et al., "ASpanFormer: Detector-free image matching with adaptive span transformer," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 20–36.

[37] Q. Wang, J. Zhang, K. Yang, K. Peng, and R. Stiefelhagen, "MatchFormer: Interleaving attention in transformers for feature matching," in *Proc. Asi. Conf. Comput. Vis. (ACCV)*, 2022, pp. 2746–2762.

[38] X. Lu and S. Du, "Raising the ceiling: Conflict-free local feature matching with dynamic view switching," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2024, pp. 256–273.

[39] Y. Wang, X. He, S. Peng, D. Tan, and X. Zhou, "Efficient LoFTR: Semi-dense local feature matching with sparse-like speed," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 21666–21675.

[40] J. Edstedt, Q. Sun, G. Bökman, M. Wadenbäck, and M. Felsberg, "RoMa: Robust dense feature matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 19790–19800.

[41] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[42] Q. Han et al., "A Siamese network via cross-domain robust feature decoupling for multi-source remote sensing image registration," *Remote Sens.*, vol. 17, no. 4, p. 646, Feb. 2025.

[43] S.-Y. Cao et al., "PCNet: A structure similarity enhancement method for multispectral and multimodal image registration," *Inf. Fusion*, vol. 94, pp. 200–214, Jun. 2023.

[44] L. Li, L. Han, and Y. Ye, "Self-supervised keypoint detection and cross-fusion matching networks for multimodal remote sensing image registration," *Remote Sens.*, vol. 14, no. 15, p. 3599, Jul. 2022.

[45] J. Chen, H. Xie, L. Zhang, J. Hu, H. Jiang, and G. Wang, "SAR and optical image registration based on deep learning with co-attention matching module," *Remote Sens.*, vol. 15, no. 15, p. 3879, Aug. 2023.

[46] L. Li, M. Liu, L. Ma, and L. Han, "Cross-modal feature description for remote sensing image matching," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102964.

[47] C. Hu, R. Zhu, X. Sun, X. Li, and D. Xiang, "Optical and SAR image registration based on pseudo-SAR image generation strategy," *Remote Sens.*, vol. 15, no. 14, p. 3528, Jul. 2023.

[48] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5728–5739.

[49] J. Leahy, S. Jabari, D. Lichti, and A. Salehitangrizi, "Enhancing cross-modal camera image and LiDAR data registration using feature-based matching," *Remote Sens.*, vol. 17, no. 3, p. 357, Jan. 2025.

[50] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[51] Y. Yao, Y. Zhang, Y. Wan, X. Liu, X. Yan, and J. Li, "Multi-modal remote sensing image matching considering co-occurrence filter," *IEEE Trans. Image Process.*, vol. 31, pp. 2584–2597, 2022.

[52] M. Yaman and S. Kalkan, "An iterative adaptive multi-modal stereo-vision method using mutual information," *J. Vis. Commun. Image Represent.*, vol. 26, pp. 115–131, Jan. 2015.

[53] K. T. Giang, S. Song, and S. Jo, "TopicFM+: Boosting accuracy and efficiency of topic-assisted feature matching," *IEEE Trans. Image Process.*, vol. 33, pp. 6016–6028, 2024.

[54] X. Jiang, J. Ren, Z. Li, X. Zhou, D. Liang, and X. Bai, "MINIMA: Modality invariant image matching," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, Mar. 2024, pp. 23059–23068.

[55] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, no. 6, pp. 726–740, 1987.

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